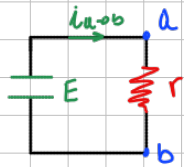


Ejemplo 1:

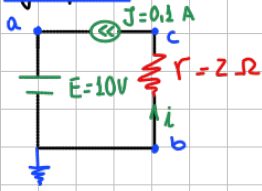


$$U_a - U_b = E$$

$$U_a - U_b = i_{a \rightarrow b} \cdot r$$

$$\left| \begin{array}{l} E = i_{a \rightarrow b} \cdot r \\ i_{a \rightarrow b} = \frac{E}{r} \end{array} \right.$$

Ejemplo 2:

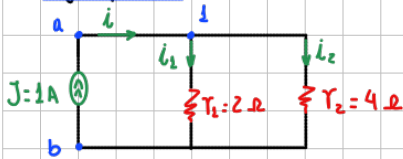


$$U_b - U_c = i \cdot r = J \cdot r \rightarrow U_{bc} = 0.2V ; U_{bc} = U_b - U_c \rightarrow U_c = -0.2V$$

$$U_a - U_b = E \rightarrow U_a = 10V ; U_b = 0V ; U_c = -0.2V$$

$$U_{ac} = U_a - U_c = 10.2V$$

Ejemplo 3:

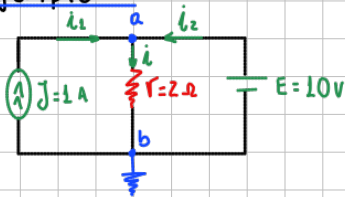


$$r_{eq} = \left(\frac{1}{2} + \frac{1}{4} \right)^{-1} = 4/3 \Omega$$

$$U_{ab} = i \cdot r_{eq} \rightarrow U_{ab} = 4/3 \Omega \cdot 1A = 4/3 V$$

$$i_1 = \frac{U_{ab}}{r_1} = 2/3 A ; i_2 = \frac{U_{ab}}{r_2} = 1/3 A$$

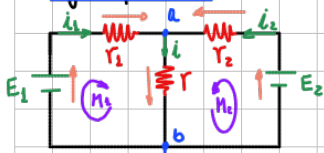
Ejemplo 4:



$$i_1 + i_2 = i \rightarrow i_1 = 1A ; i_2 = i - i_1 = 4A$$

$$U_a - U_b = i \cdot r = E \rightarrow i = \frac{E}{r} = 5A$$

Ejemplo 5:



$$E_1 = 10 ; E_2 = 12$$

$$r_1 = 1 ; r_2 = 2 ; r = 10$$

$$\text{Nodo a: } i_1 + i_2 = i$$

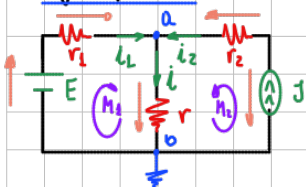
$$\text{Malla 1: } E_1 = i_1 \cdot r_1 + i \cdot r$$

$$\text{Malla 2: } E_2 = i_2 \cdot r_2 + i \cdot r$$

$$\begin{bmatrix} i_1 & i_2 & i \\ 1 & 1 & -1 \\ 1 & 0 & 10 \\ 0 & 2 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 12 \end{bmatrix}$$

$$i_1 = 0 ; i_2 = 1A ; i_3 = 1A$$

Ejemplo 6:



$$E = 10V ; J = 0.1A$$

$$r_1 = 1\Omega ; r_2 = 2\Omega ; r = 10\Omega$$

$$\text{Nodo a: } i_1 + i_2 - i = 0$$

$$\text{Malla 1: } i_1 \cdot r_1 + i \cdot r = E$$

$$\text{Malla 2: } i_2 \cdot r_2 + i \cdot r - U_3 = 0$$

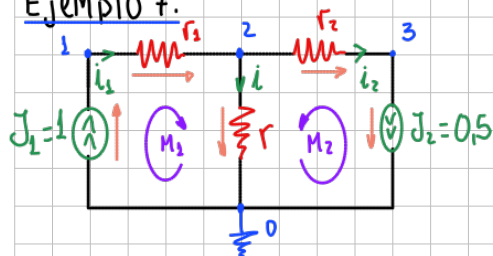
$$\begin{bmatrix} i_1 & i & U_3 \\ 1 & -1 & 0 \\ 1 & 10 & 0 \\ 0 & 10 & -1 \end{bmatrix} \begin{bmatrix} i_1 \\ i \\ U_3 \end{bmatrix} = \begin{bmatrix} -0.1 \\ 10 \\ -0.2 \end{bmatrix}$$

$$i_1 = 0.8182 A$$

$$i = 0.9182$$

$$U_3 = 9.3818 V$$

Ejemplo 7:



Datos: $r_1 = 1 \Omega$; $r_2 = 2 \Omega$; $r = 10$; $J_1 = 1 \text{ A}$; $J_2 = 0.5 \text{ A}$

$$\sum E_i = \sum U_i \rightarrow \begin{cases} U_{J1} + i_1 \cdot r_1 + i \cdot r = 0 \\ i \cdot r - i_2 \cdot r_2 - U_{J2} = 0 \end{cases} \quad \begin{cases} i_1 = J_1 = 1 \text{ A} \\ i_2 = J_2 = 0.5 \text{ A} \end{cases}$$

$$\sum i_{\text{ent}} = \sum i_{\text{sal}} \rightarrow i_1 - i_2 - i = 0$$

$$i \quad U_{J1} \quad U_{J2}$$

$$\begin{bmatrix} r & 1 & 0 \\ r & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ U_{J1} \\ U_{J2} \end{bmatrix} = \begin{bmatrix} -i_1 r_1 \\ i_2 r_2 \\ J_1 - J_2 \end{bmatrix}$$

$$i = 0.5 \text{ A}$$

$$U_{J1} = -6 \text{ V}$$

$$U_{J2} = 4 \text{ V}$$

$$\begin{cases} i \cdot r = U_2 - U_0 = 5 \text{ V} = U_2 \\ U_{J1} = U_0 - U_1 = 6 \text{ V} = U_2 \\ U_{J2} = U_3 - U_0 = 4 \text{ V} = U_3 \end{cases} \quad \begin{cases} i_2 \cdot r_2 = U_2 - U_3 \\ 1 = 1 \\ i_1 \cdot r_1 = U_1 - U_2 \\ 1 = 1 \end{cases}$$

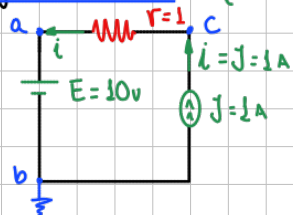
Otro Metodo:

$$\begin{bmatrix} \frac{1}{r_1} & -\frac{1}{r_1} & 0 \\ -\frac{1}{r_1} & \frac{1}{r_1} + \frac{1}{r} + \frac{1}{r_2} & -\frac{1}{r_2} \\ 0 & -\frac{1}{r_2} & \frac{1}{r_2} \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} J_1 \\ 0 \\ -J_2 \end{bmatrix}$$

$$U_1 = 6; U_2 = 5; U_3 = 4$$

$$\begin{cases} i \cdot r = U_2 - U_0 \\ i = \frac{U_2}{r} = 0.5 \text{ A} \end{cases}$$

Ejercicio 9: (Balance de Potencia.)



$$U_{ab} = E = 10 \text{ V} = U_a - U_b$$

$$P_{cE} = U_{ab} \cdot i_{a \rightarrow b} = 10 \text{ W}$$

$$P_{cR} = i_{c \rightarrow b} \cdot r = 1 \text{ W}$$

$$P_{gJ} = U_{cb} \cdot i_{b \rightarrow c} = 11 \text{ W}$$

$$\sum P_g = \sum P_c$$

$$11 \text{ W} = 10 \text{ W} + 1 \text{ W}$$

"Las Fuentes consumen potencia cuando la corriente va de un potencial mayor a uno menor y viceversa."

Ejercicio 10:

Datos: $r_{12} = 1.5$; $r_{23} = 1.5$; $r_{13} = 2$; $r_{20} = 10$; $r_{10} = 0.1$

$$E = 100; J = 0.1$$

$$G \cdot U = J$$

$$\begin{bmatrix} \frac{1}{r_{10}} + \frac{1}{r_{12}} + \frac{1}{r_{13}} & -\frac{1}{r_{12}} & -\frac{1}{r_{13}} \\ -\frac{1}{r_{12}} & \frac{1}{r_{12}} + \frac{1}{r_{20}} + \frac{1}{r_{23}} & -\frac{1}{r_{23}} \\ -\frac{1}{r_{13}} & -\frac{1}{r_{23}} & \frac{1}{r_{23}} + \frac{1}{r_{13}} \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} E/r_{10} \\ 0 \\ -J \end{bmatrix}$$

$$\begin{cases} U_1 = 99.094 \\ U_2 = 89.623 \\ U_3 = 93.596 \end{cases}$$

$$i_{10} = \frac{E - U_1}{r_{10}} = 9.0623 \text{ A}$$

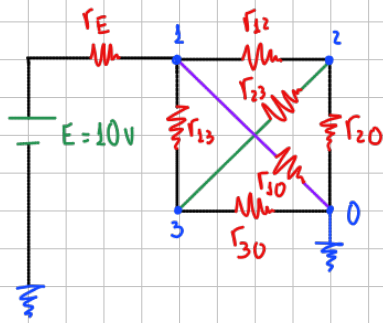
$$i_{12} = \frac{U_1 - U_2}{r_{12}} = 6.3136 \text{ A}$$

$$i_{13} = \frac{U_1 - U_3}{r_{13}} = 2.7487 \text{ A}$$

$$i_{23} = \frac{U_2 - U_3}{r_{23}} = -2.6487 \text{ A}$$

$$i_{20} = \frac{U_2}{r_{20}} = 8.9623 \text{ A}$$

Ejercicio 11:



Datos: $r_E = 1$; $r_{12} = 1$; $r_{13} = 5$; $r_{10} = 12$;
 $r_{20} = 6$; $r_{23} = 10$ / $r_{30} = 1$ / $E = 10V$

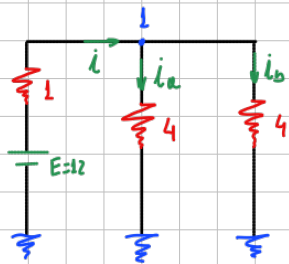
$$\begin{bmatrix} \frac{1}{r_E} + \frac{1}{r_{12}} + \frac{1}{r_{13}} + \frac{1}{r_{10}} & -\frac{1}{r_{12}} & -\frac{1}{r_{13}} \\ -\frac{1}{r_{12}} & \frac{1}{r_{12}} + \frac{1}{r_{20}} + \frac{1}{r_{23}} & -\frac{1}{r_{23}} \\ -\frac{1}{r_{13}} & -\frac{1}{r_{23}} & \frac{1}{r_{13}} + \frac{1}{r_{23}} + \frac{1}{r_{30}} \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} E/r_E \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{cases} U_1 = 6,9753 \\ U_2 = 5,6257 \\ U_3 = 1,5059 \end{cases}$$

$$i_{12} = \frac{U_1 - U_2}{r_{12}} = 1,3496 ; i_{13} = \frac{U_1 - U_3}{r_{13}} = 1,0939 ; i_{10} = \frac{U_1}{r_{10}} = 0,5813 ; i_E = \frac{E - U_1}{r_E} = 3,0247$$

$$i_{23} = \frac{U_2 - U_3}{r_{23}} = 0,4120 ; i_{20} = \frac{U_2}{r_{20}} = 0,9376$$

$$i_{30} = \frac{U_3}{r_{30}} = 1,5059$$

Ejercicio 12:



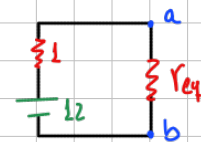
Metodo 1:

$$[G] \cdot [U] = [J]$$

$$\left(1 + \frac{1}{4} + \frac{1}{4}\right) \cdot U_1 = \frac{12}{1} \rightarrow U_1 = 8V$$

$$i = \frac{12 - 8}{1} = 4A ; i_a = \frac{8}{4} = 2A ; i_b = \frac{8}{4} = 2A$$

Metodo 2:



$$r_{eq} = \frac{4 \cdot 4}{4 + 4} = 2$$

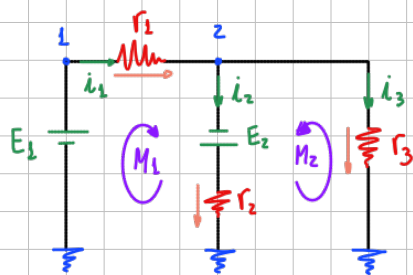
$$r_T = 3$$

$$i = \frac{12}{3} = 4A$$

$$U_{ab} = i \cdot r_{eq} = 8V$$

$$i_a = \frac{8}{4} = 2A ; i_b = \frac{8}{4} = 2A$$

Ejercicio 13:



Metodo 1: (Kirchoff)

$$i_1 \cdot r_1 + i_2 \cdot r_2 = E_1 - E_2$$

$$i_2 \cdot r_2 - i_3 \cdot r_3 = E_2$$

$$i_1 - i_2 - i_3 = 0$$

$$\begin{bmatrix} r_1 & r_2 & 0 \\ 0 & r_2 & -r_3 \\ 1 & -1 & -1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} E_1 + E_2 \\ E_2 \\ 0 \end{bmatrix} \rightarrow \begin{cases} i_1 = 4,6A \\ i_2 = 4,4A \\ i_3 = 0,2A \end{cases}$$

$$U_1 = 10V ; U_2 = 10 - i_1 \cdot r_1 = 0,8V = i_2 \cdot r_2 - E_2$$

Datos:

$$r_1 = 2 ; r_2 = 2 ; r_3 = 4$$

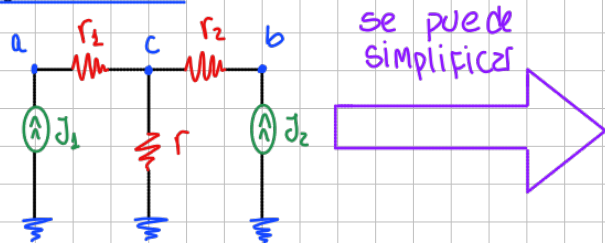
$$E_1 = 10 ; E_2 = 8$$

Metodo 2: (Potenciales de Nodos)

$$\left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}\right) \cdot U_2 = \left(\frac{E_1}{r_1} - \frac{E_2}{r_2}\right) \rightarrow U_2 = 0,8V$$

$$i_1 = \frac{E_1 - U_1}{r_1} = 4,6A ; i_2 = \frac{E_2 + U_2}{r_2} = 4,4A ; i_3 = \frac{U_2}{r_3} = 0,2A$$

Ejercicio 15:



$$\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \cdot \begin{bmatrix} U_a \\ U_b \\ U_c \end{bmatrix} = \begin{bmatrix} J_{11} \\ J_{12} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{r_1} & 0 \\ 0 & \frac{1}{r_2} \\ -\frac{1}{r_1} & -\frac{1}{r_2} \end{bmatrix} \cdot \begin{bmatrix} U_a \\ U_b \\ U_c \end{bmatrix} = \begin{bmatrix} J_1 \\ J_2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \cdot \begin{bmatrix} U_{11} \\ U_{21} \end{bmatrix} = \begin{bmatrix} J_{11} \\ 0 \end{bmatrix}$$

$$G_{11} \cdot U_{11} + G_{12} \cdot U_{21} = J_{11}$$

$$G_{21} \cdot U_{11} + G_{22} \cdot U_{21} = 0$$

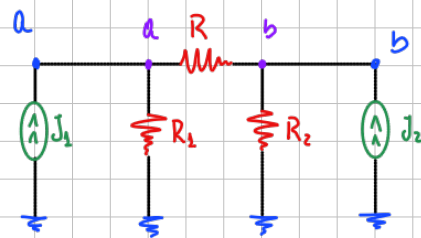
$$G_{22}^{-1} \cdot G_{22} \cdot U_{21} = -G_{22}^{-1} \cdot G_{21} \cdot U_{11}$$

$$U_{21} = -G_{22}^{-1} \cdot G_{21} \cdot U_{11}$$

$$G_{11} \cdot U_{11} + G_{12} \cdot (-G_{22}^{-1} \cdot G_{21} \cdot U_{11}) = J_{11}$$

$$G_{11} \cdot U_{11} \cdot U_{11}^{-1} + G_{12} \cdot (-G_{22}^{-1} \cdot G_{21}) \cdot U_{11} \cdot U_{11}^{-1} = J_{11} \cdot U_{11}^{-1}$$

$$G_{11} - G_{12} \cdot G_{22}^{-1} \cdot G_{21} = G^{eq}$$



$$\underbrace{\begin{bmatrix} \frac{1}{R} + \frac{1}{R_1} & -\frac{1}{R} \\ -\frac{1}{R} & \frac{1}{R} + \frac{1}{R_2} \end{bmatrix}}_{G'} \cdot \underbrace{\begin{bmatrix} U_a \\ U_b \end{bmatrix}}_{U'} = \underbrace{\begin{bmatrix} J_1 \\ J_2 \end{bmatrix}}_{J'}$$

Para encontrar los valores de "R" que verifican la igualdad, se tiene que cumplir:

$$G^{eq} = G'$$

Trabajando simbólicamente se obtiene:

$$R = \frac{r_1 \cdot r_2 + r_1 \cdot r + r_2 \cdot r}{r}$$

$$R_1 = \frac{r_1 \cdot r_2 + r_1 \cdot r + r_2 \cdot r}{r_2}$$

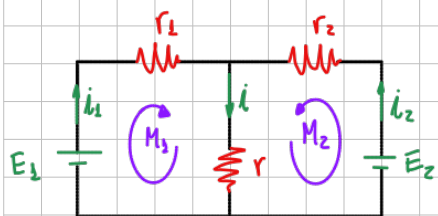
$$R_2 = \frac{r_1 \cdot r_2 + r_1 \cdot r + r_2 \cdot r}{r_1}$$

¡ Dichos resultados son para este circuito en específico !

Ejercicio 16:

ITENCIDADES DE MALLA

$$[R] \cdot [I] = [V] \quad (I: \text{corriente de Malla})$$



• i_1 : Corriente de Malla 1

• i_2 : Corriente de Malla 2

• i, i_1, i_2 : Corriente de rama

Datos:

$$E_1 = 10V; E_2 = 12V;$$

$$r_1 = 1\Omega; r_2 = 1,5\Omega; r = 10\Omega$$

$$\begin{bmatrix} r_1 + r & -r \\ -r & r_2 + r \end{bmatrix} \cdot \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} E_1 \\ E_2 \end{bmatrix}$$

$$i_1 = -0,1887A; i_2 = 1,2075A$$

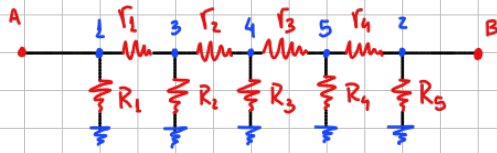
$$i = i_1 + i_2 = 1,0189A$$

Ejercicio 17:

Datos:

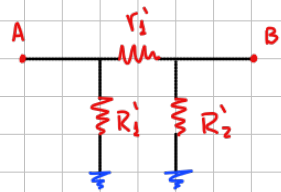
$$r_1 = r_2 = 1; r_3 = r_4 = 1,2$$

$$R_1 = R_2 = 10; R_3 = R_4 = R_5 = 15$$



$$G_{11} \begin{bmatrix} \frac{1}{r_1} + \frac{1}{R_1} & 0 & -\frac{1}{r_1} & 0 & 0 \\ 0 & \frac{1}{r_2} + \frac{1}{R_2} & 0 & 0 & -\frac{1}{r_2} \\ -\frac{1}{r_1} & 0 & \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{R_3} & -\frac{1}{r_2} & 0 \\ 0 & 0 & -\frac{1}{r_2} & \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{R_4} & -\frac{1}{r_3} \\ 0 & -\frac{1}{r_2} & 0 & -\frac{1}{r_3} & \frac{1}{r_3} + \frac{1}{r_4} + \frac{1}{R_5} \end{bmatrix} \cdot \begin{bmatrix} U_A \\ U_B \\ U_3 \\ U_4 \\ U_5 \end{bmatrix} = \begin{bmatrix} J_A \\ J_B \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$G^{eq} = G_{11} - G_{12} \cdot G_{22}^{-1} \cdot G_{21}$$



$$\begin{bmatrix} 0,4022 & -0,1865 \\ -0,1865 & 0,3425 \end{bmatrix} \cdot \begin{bmatrix} U_A \\ U_B \end{bmatrix} = \begin{bmatrix} J_A \\ J_B \end{bmatrix}$$

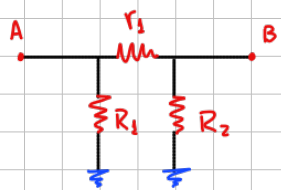
$$\begin{aligned} & \cdot \frac{-1}{r_1} = -0,1865 \rightarrow r_1' = 5,3619 \Omega \\ & \cdot \frac{1}{r_1'} + \frac{1}{R_1} = 0,4022 \rightarrow R_1' = 4,6361 \Omega \\ & \cdot \frac{1}{r_1'} + \frac{1}{R_2} = 0,3425 \rightarrow R_2' = 6,4095 \Omega \end{aligned}$$

• Verificación: "Inyecto un J_A y J_B en G y G^{eq} y deberíamos obtener el mismo U_A y U_B "

• Tomando $J_1 = 1$ y $J_2 = 1$:

$$\begin{bmatrix} 5,1367 \\ 5,7165 \\ 4,6504 \\ 4,6291 \\ 4,9738 \end{bmatrix} = \begin{bmatrix} 5,1367 \\ 5,7165 \end{bmatrix}$$

Ejercicio 18: "Determinación de circuito equivalente mediante Mediciones."



1) Aplico diferentes J_A y J_B (Vectores independientes y los necesarios)

$$[G^{eq}] \cdot [U_A] = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Mido Aplico

$$[G^{eq}] \cdot [U_B] = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Mido Aplico

* En este caso para no medir uso el G^{eq} del ejercicio 17

$$[G]^{-1} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5,1374 \\ 5,7171 \end{bmatrix} = [U_A]; \quad [G]^{-1} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3,3262 \\ 1,8112 \end{bmatrix} = [U_B]$$

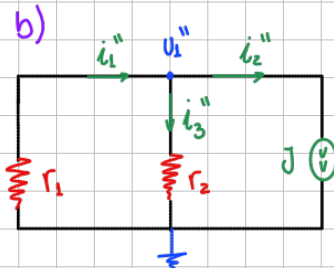
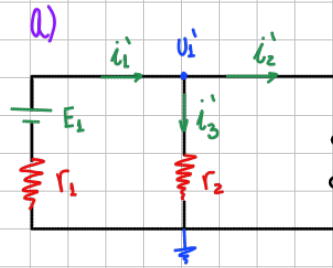
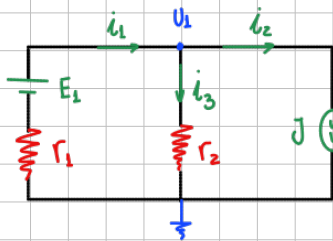
2) Nos aprovechamos que por ser una transformación Matricial, es una trans. Lineal:

$$[G^{eq}] \cdot \begin{bmatrix} U_A \\ U_B \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \rightarrow [G^{eq}] = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 5,1374 & 3,3262 \\ 5,7171 & 1,8112 \end{bmatrix}$$

$$\begin{aligned} r_1' &= 5,3619 \\ R_1' &= 4,6361 \\ R_2' &= 6,4103 \end{aligned}$$

$$[G^{eq}] = \begin{bmatrix} 0,4022 & -0,1865 \\ -0,1865 & 0,3425 \end{bmatrix}$$

Ejercicio 19:



Datos:

$$E_1 = 100 \text{ V}$$

$$J = 5 \text{ A}$$

$$r_1 = 1; r_2 = 20$$

Aplicando Superposición

$$i_1 = 9,5238 \text{ A}$$

$$i_2 = 5 \text{ A}$$

$$i_3 = 4,5239 \text{ A}$$

$$U_3 = 90,4762 \text{ V}$$

$$i_2' = 0; i_1' = i_3'$$

$$i_1' \cdot r_2 + i_1' \cdot r_1 = E_1 \rightarrow i_1' = 4,7619 \text{ A}; U_1' = E_1 - i_1' \cdot r_1$$

$$U_1' = 95,2381 \text{ V}$$

$$\left[\frac{1}{r_1} + \frac{1}{r_2} \right] \cdot [U_1''] = [-J] \rightarrow U_1'' = -4,7619 \text{ V}$$

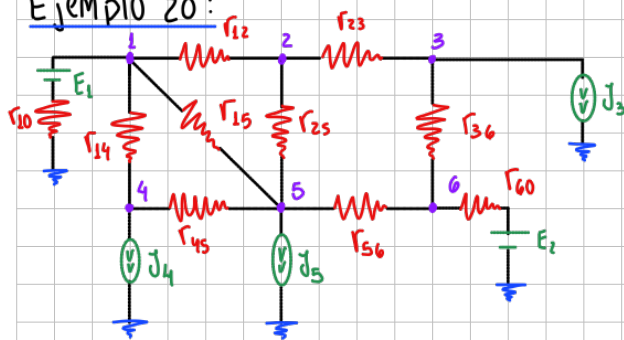
$$i_1'' = \frac{-U_1''}{r_1} = 4,7619; i_2'' = 5; i_3'' = \frac{U_1''}{r_2} = -0,2380 \text{ A}$$

Verificación: (Kirchoff)

$$i_1 \cdot r_1 + i_3 \cdot r_2 = E \rightarrow i_1 = 9,5238 \text{ A}; i_2 = 5 \text{ A}; i_3 = 4,5238; U_3 = E_1 - i_1 \cdot r_1 = 90,4762 \text{ V}$$

$$i_1 - i_3 = 5 \text{ A}$$

Ejemplo 20:



Datos:

$$r_{12} = 1; r_{23} = 3; r_{36} = 1; r_{56} = 1; r_{45} = 1; r_{14} = 1; r_{25} = 2; r_{35} = 2$$

$$r_{10} = 0,5; r_{60} = 0,5$$

$$E_1 = ?; E_2 = ? / J_4 = J_3 = J_5 = 1 \text{ A}$$

Matriz Nodal:

$$1) \text{ Planteo } \rightarrow [G] \cdot [U] = [B] \rightarrow [U] = [G]^{-1} \cdot [B]$$

2) Matriz Excitación B:

$$[B] = K_1 \cdot b_1 + K_2 \cdot b_2 + b_0; [b_1] = \begin{bmatrix} E_1/r_{10} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}; b_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ E_2/r_{60} \end{bmatrix}; b_0 = \begin{bmatrix} 0 \\ 0 \\ -J_3 \\ -J_4 \\ -J_5 \\ 0 \end{bmatrix}$$

3) Planteo la solución:

$$[U] = \underbrace{[G]^{-1} \cdot K_1}_{[a]} \cdot \underbrace{[b_1]}_{[c]} + \underbrace{[G]^{-1} \cdot K_2}_{[c]} \cdot \underbrace{[b_2]}_{[d]} + \underbrace{[G]^{-1} \cdot b_0}_{[d]}$$

$$[U] = [a] \cdot K_1 + [c] \cdot K_2 + [d]$$

4) Aplico la restricción $95 < U_i < 105$

$$95 < a_i \cdot K_1 + c_i \cdot K_2 + d_i < 105$$

$$95 - d_i < a_i \cdot K_1 + c_i \cdot K_2 < 105 - d_i \longrightarrow \text{Conjunto solución}$$

¡ En Octave user $\longrightarrow$ • ejercicio_20
• ejercicio_20.CHATGPT_pro

5) Optimización del costo.

Suponer: $C = Z \cdot P_{E_1} + P_{E_2} \longrightarrow P_{E_1} = E_1 \cdot I_1 ; P_{E_2} = E_2 \cdot I_6$

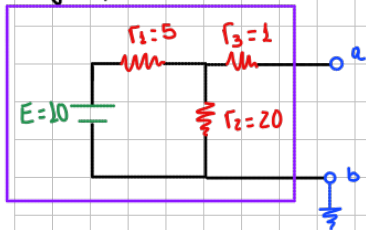
• corriente: $I_1 = \frac{K_1 E_1 - U_1}{r_{10}} ; I_6 = \frac{K_2 E_2 - U_6}{r_{60}} ; U_1 = a_1 \cdot K_1 + c_1 \cdot K_2 + d_1 ; U_6 = a_6 \cdot K_1 + c_6 \cdot K_2 + d_6$

$$C = Z \cdot E_1 \cdot \frac{K_1 E_1 - U_1}{r_{10}} + E_2 \cdot \frac{K_2 E_2 - U_6}{r_{60}}$$

$$C = Z \cdot E_1 \cdot \frac{K_1 \cdot E_1 - (a_1 \cdot K_1 + c_1 \cdot K_2 + d_1)}{r_{10}} + E_2 \cdot \frac{K_2 \cdot E_2 - (a_6 \cdot K_1 + c_6 \cdot K_2 + d_6)}{r_{60}}$$

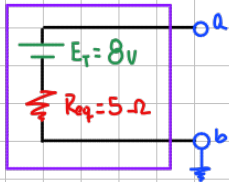
$$\text{Min } C ; \text{ sujeto a: } 95 - d_i < a_i \cdot K_1 + c_i \cdot K_2 < 105 - d_i$$

Ejemplo 21: "Teorema del dipolo activo, Thevelin y Norton"



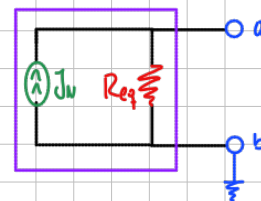
1) Thevelin: 1) $R_{eq} = \frac{20 \cdot 5}{20+5} + 1 = 5 \longrightarrow R_{eq} = 5 \Omega$

2) $i \cdot 5 + i \cdot 20 = 10 \longrightarrow 10V - 0,45 = 8V \longrightarrow V_{ab} = 8V$
 $i = 0,4A$



2) Norton: 1) $R_{eq} = \frac{20 \cdot 5}{20+5} + 1 = 5 \longrightarrow R_{eq} = 5 \Omega$

2) $\frac{8V}{5\Omega} = 1,6A \longrightarrow J_N = 1,6A$



Pasos Thevelin:

1) Elimino las Fuentes y calculo R_{eq} .

2) calcular la V_{ab}

Pasos Norton:

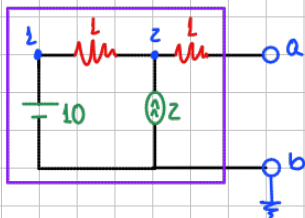
1) $R_{eq} \longrightarrow$ Thevelin

2) $J_N = \frac{V_{th}}{R_{eq}}$: cortocir

VERIFICACION:

$$\frac{V_{th}}{J_N} = R_{eq}$$

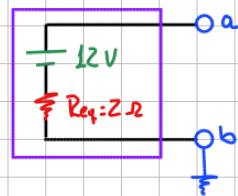
Ejemplo 22:



Thevenin

$$1) 1 + 1 = 2 \rightarrow R_{eq} = 2 \Omega$$

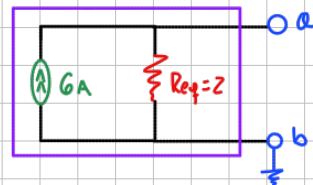
$$2) U_{21} = 1 \cdot 2 = 2V \rightarrow V_{ab} = 10V + 2V = 12V$$



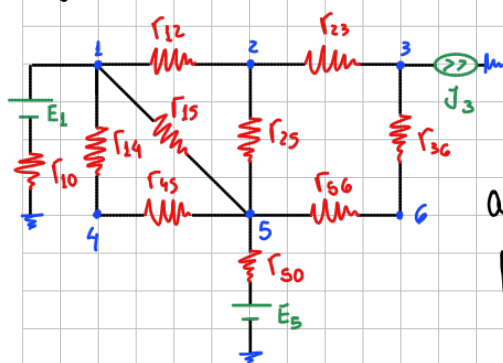
Norton:

$$1) 1 + 1 = 2 \rightarrow R_{eq} = 2 \Omega$$

$$2) \left(\frac{1}{2}\right) \cdot U_{th} = J_N \rightarrow J_N = 6A$$



Ejemplo 23:



Datos: $E_1 = 100V$; $E_5 = 105V$; $J_3 = 2A$

$$r_{12} = 1; r_{14} = 1; r_{15} = 2; r_{23} = 2; r_{25} = 1; r_{36} = 3; r_{56} = 1; r_{45} = 1$$

$$r_{10} = 0,5; r_{50} = 0,1$$

a) ARMO $[G] \cdot [U] = [J]$

$$\begin{bmatrix} 4,5 & -1 & 0 & -1 & -0,5 & 0 \\ -1 & 2,5 & -0,5 & 0 & -1 & 0 \\ 0 & -0,5 & 0,833 & 0 & 0 & -0,333 \\ -1 & 0 & 0 & 2 & -1 & 0 \\ -0,5 & -1 & 0 & -1 & 13,5 & -1 \\ 0 & 0 & -0,33 & 0 & -1 & 1,33 \end{bmatrix} \begin{bmatrix} 200 \\ 0 \\ -2 \\ 0 \\ 1050 \\ 0 \end{bmatrix}$$

b) ARMO U_{th} : $\rightarrow [U_{th}] = [G]^{-1} \cdot [J]$

$$\begin{bmatrix} 101,76 \\ 102,59 \\ 100,54 \\ 103,10 \\ 104,45 \\ 103,47 \end{bmatrix} = [U_{th}]$$

c) calculo de R_i

$$[G] \cdot \begin{bmatrix} 0 \\ \vdots \\ U_i \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow R_i = U_i \text{ (simulo inyectar un } J_i = 1A \text{ por eso } R_i = U_i)$$

$$R_i = \frac{U_i}{1A}$$

$$R_1 = 0,3; R_2 = 0,572; R_3 = 1,628; R_4 = 0,618;$$

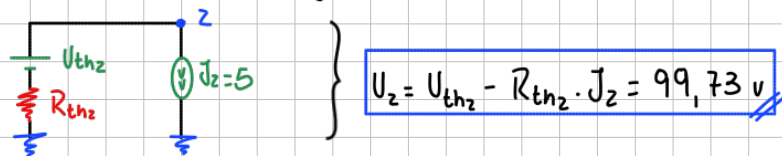
$$R_5 = 0,092; R_6 = 0,932$$

d) $J_{Ni} = \frac{U_{thi}}{R_{thi}}; P_{cc} = U_{thi} \cdot J_{Ni} / J_N = J_{cc}$

$$J_{N1} = 339,2; J_{N2} = 179,357; J_{N3} = 61,759; J_{N4} = 166,835; J_{N5} = 1135,304; J_{N6} = 111,021$$

$$P_{cc1} = 34517; P_{cc2} = 18401; P_{cc3} = 6209; P_{cc4} = 17201; P_{cc5} = 118582; P_{cc6} = 11488$$

Ejemplo 24: En el ejercicio anterior se conecta un: $J_2 = 5 \text{ A}$

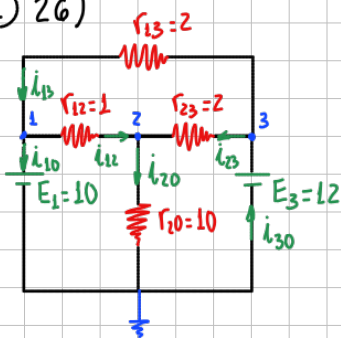


Ejemplo 25: Verificar el ej 24 usando $[W] = [G]^{-1} \cdot [J]$

$$[G]^{-1} \cdot [J] = \begin{bmatrix} 100,96 \\ 99,732 \\ 98,524 \\ 102,534 \\ 104,108 \\ 102,712 \end{bmatrix}$$

Repaso Corriente Continua:

① 26)



Primer Metodo:

$$\left(\frac{1}{r_{12}} + \frac{1}{r_{23}} + \frac{1}{r_{20}} \right) \cdot U_2 = \left(\frac{E_1}{r_{12}} + \frac{E_2}{r_{23}} \right) \rightarrow U_2 = 10 ; U_1 = 10 ; U_3 = 12$$

$$i_{12} = \frac{E_1 - U_2}{r_{12}} = 0 ; i_{23} = \frac{U_3 - U_2}{r_{23}} = 1 ; i_{13} = \frac{U_3 - U_1}{r_{13}} = 1 ; i_{20} = \frac{U_2}{r_{20}} = 1$$

$$i_{10} = i_{13} = 1 ; i_{30} = i_{23} + i_{13} = 2$$

Segundo Metodo: $r_F \downarrow \downarrow = 0,00000001$

$$\begin{bmatrix} \frac{1}{r_F} + \frac{1}{r_{12}} + \frac{1}{r_{13}} & -\frac{1}{r_{12}} & -\frac{1}{r_{13}} \\ -\frac{1}{r_{12}} & \frac{1}{r_{12}} + \frac{1}{r_{23}} + \frac{1}{r_{20}} & -\frac{1}{r_{23}} \\ -\frac{1}{r_{13}} & -\frac{1}{r_{23}} & \frac{1}{r_{23}} + \frac{1}{r_{13}} + \frac{1}{r_F} \end{bmatrix} \cdot \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} E_1/r_F \\ 0 \\ E_3/r_F \end{bmatrix} \rightarrow U_1 = 10 ; U_2 = 10 ; U_3 = 12$$

